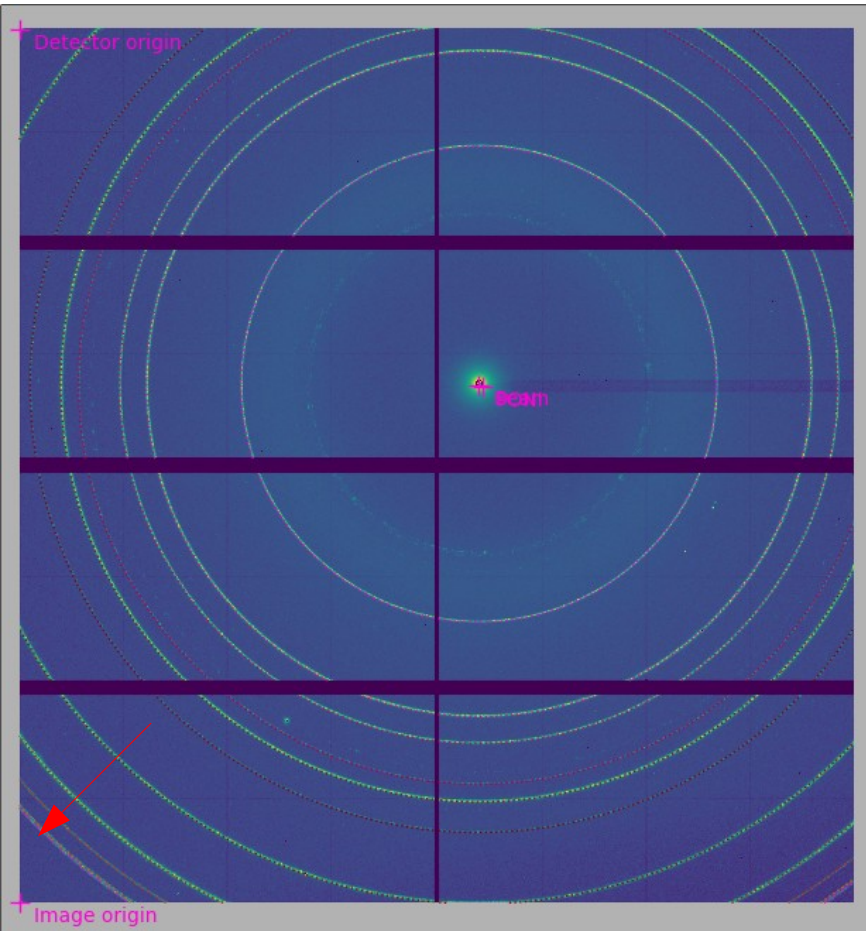


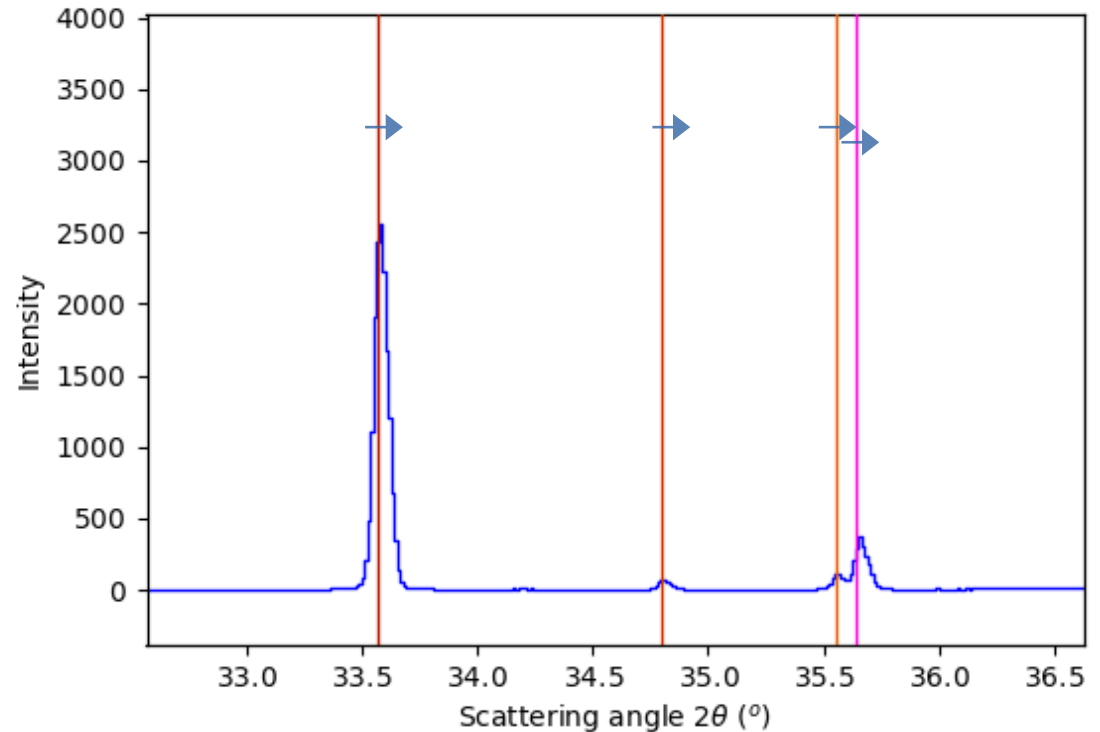
Parallax

A journey in sensor physics

Effect on ID13

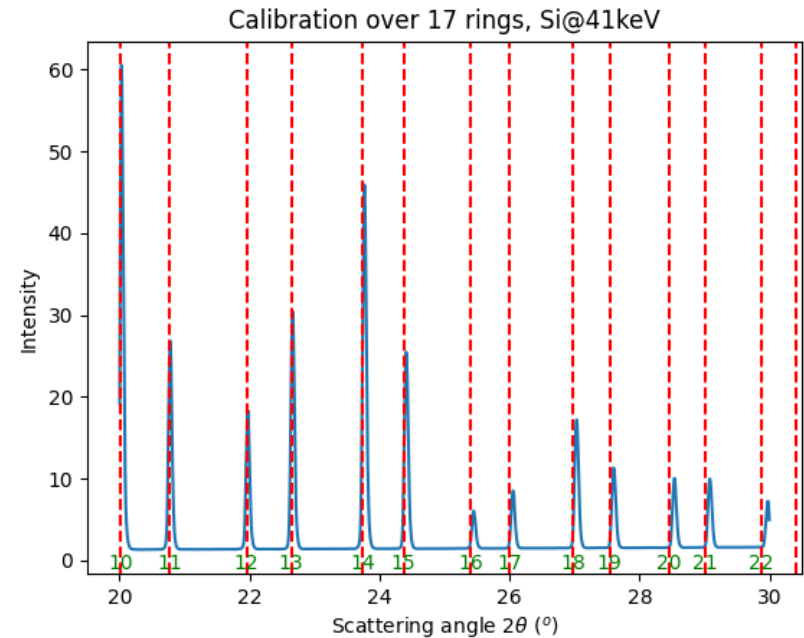


- Eiger 4M detector 75 μ m pixel size
- Silicon sensor 450 μ m thick
- Sample : alpha Al_2O_3 @13,4keV

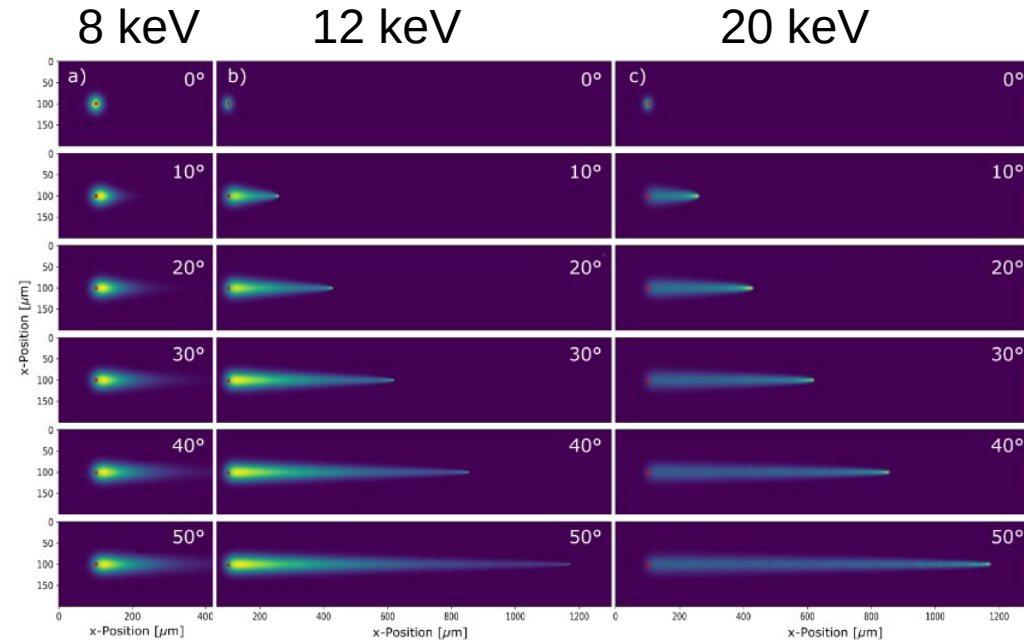
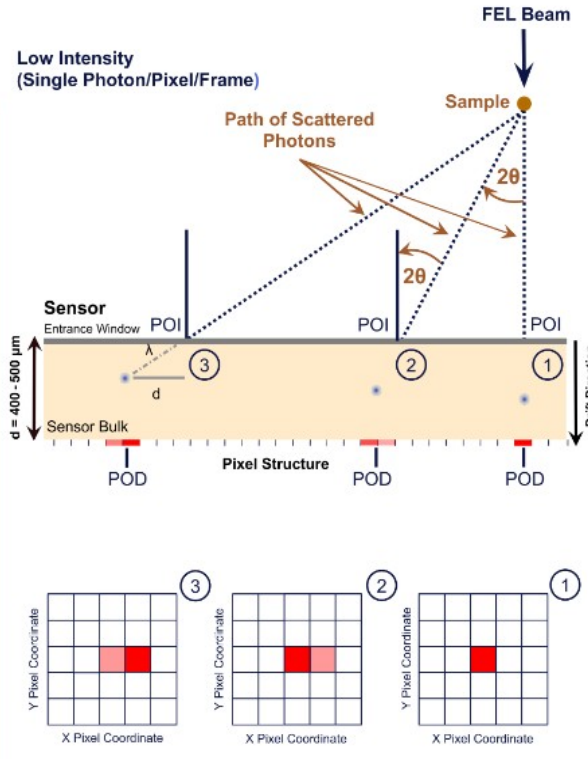
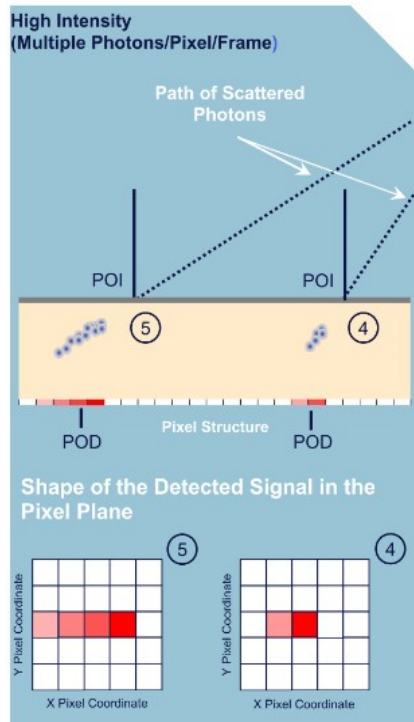


Effect on ID11

- Eiger2 4M detector, 75 μ m pixel size
- CdTe sensor 750 μ m,
- sample: Si@41keV

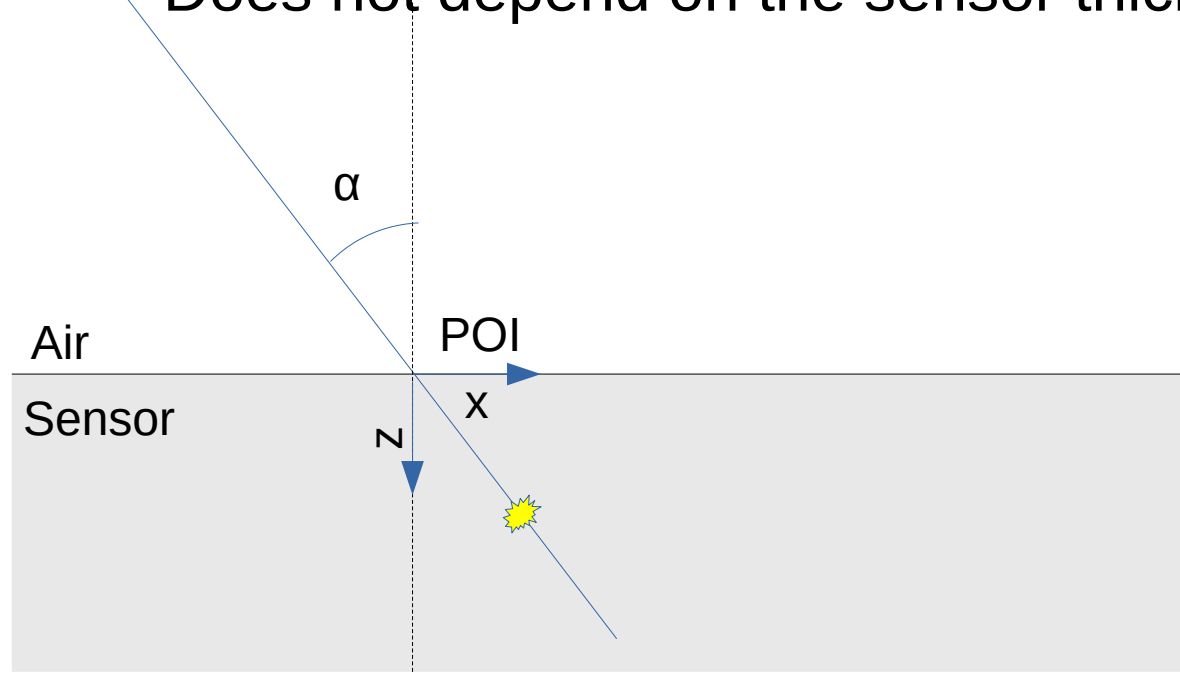


Parallax in photon detection



Thick sensor approximation

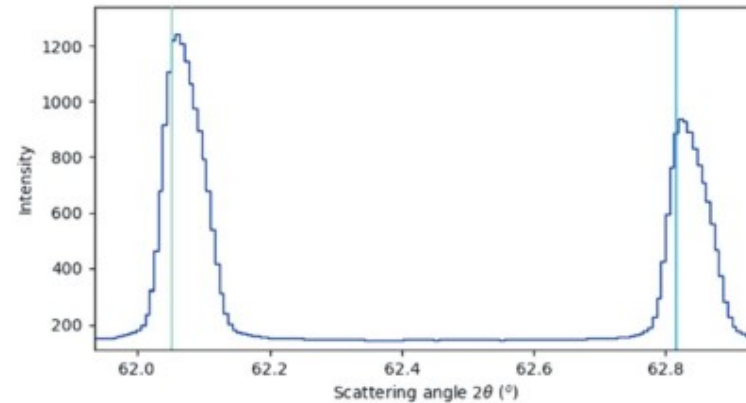
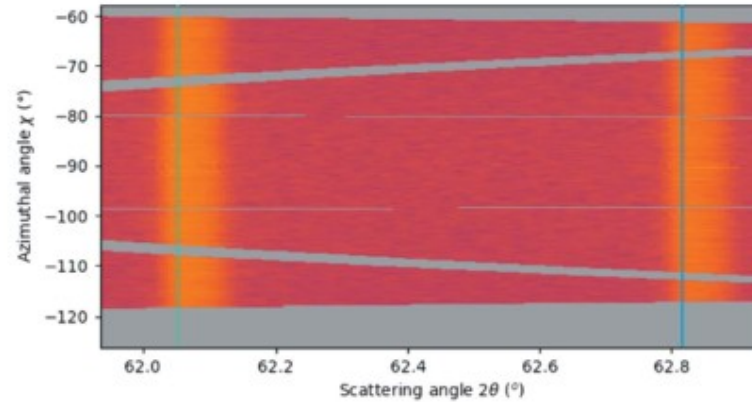
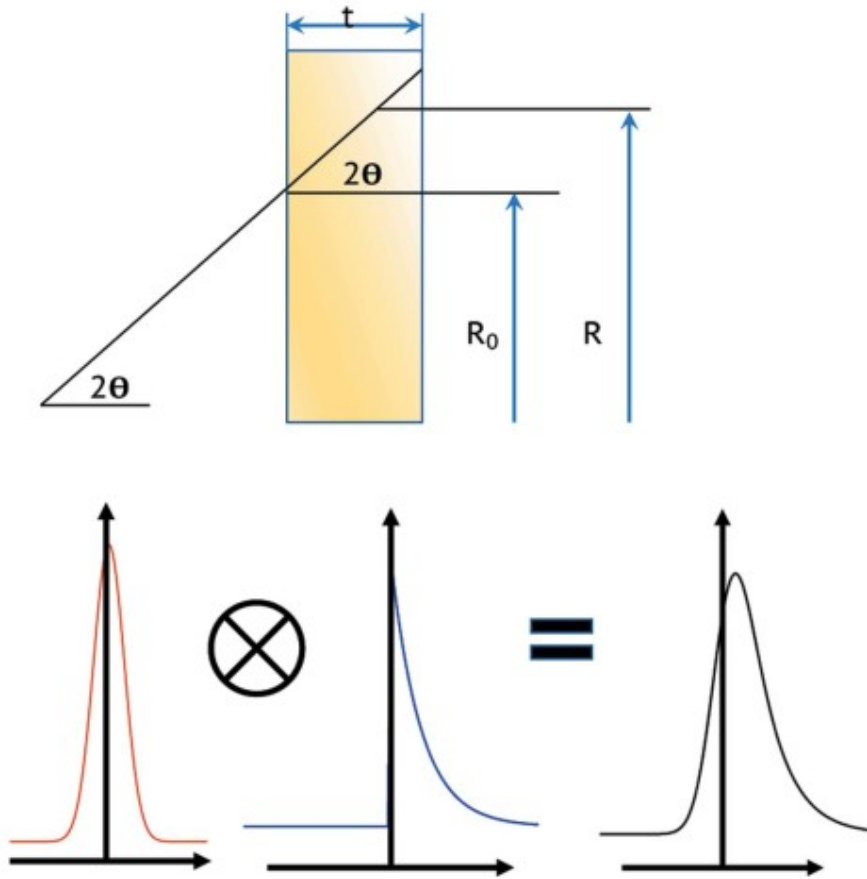
- All photons are caught
- Does not depend on the sensor thickness !



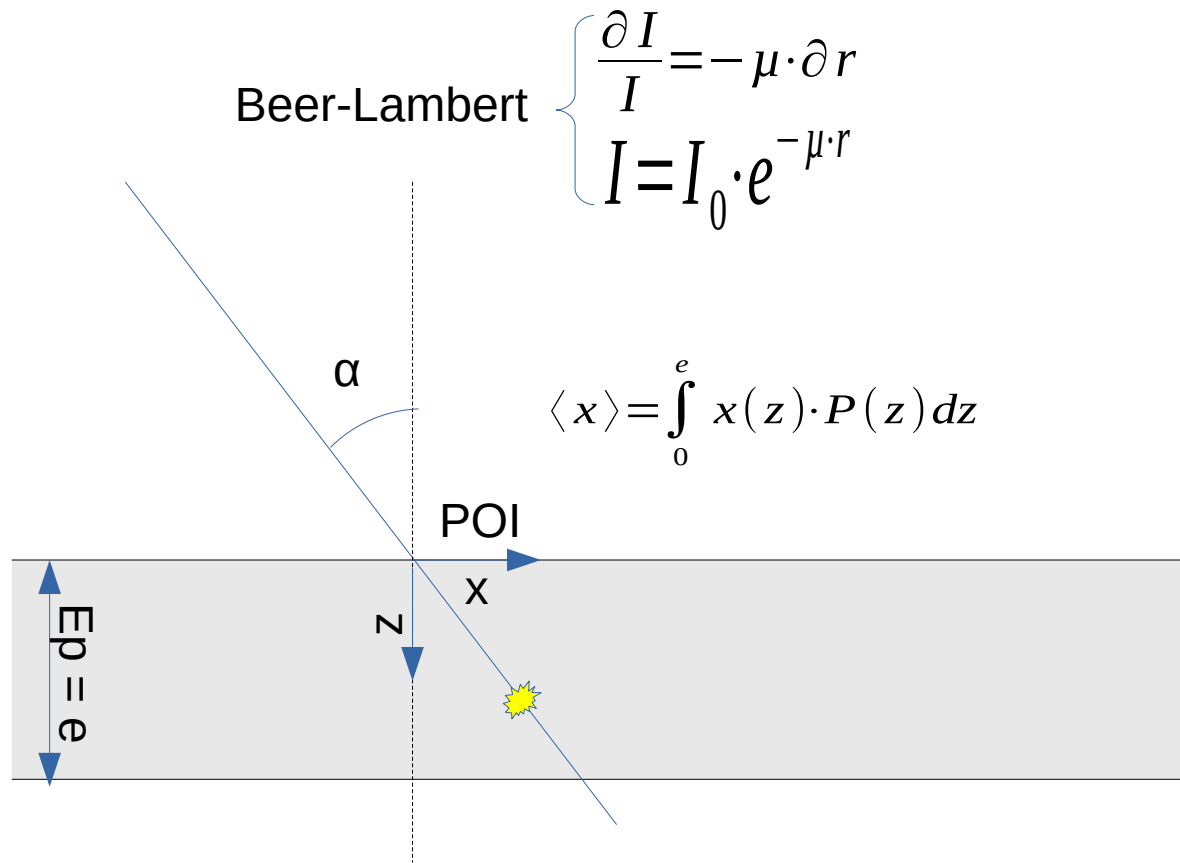
$$I = I_0 \cdot e^{-\mu \cdot r}$$

$$\Delta x = \frac{\sin(\alpha)}{\mu}$$

Thin-sensor convoluted with beam profile



Thin-sensor without beam profile



$$x(z) = z \cdot \tan(\alpha)$$

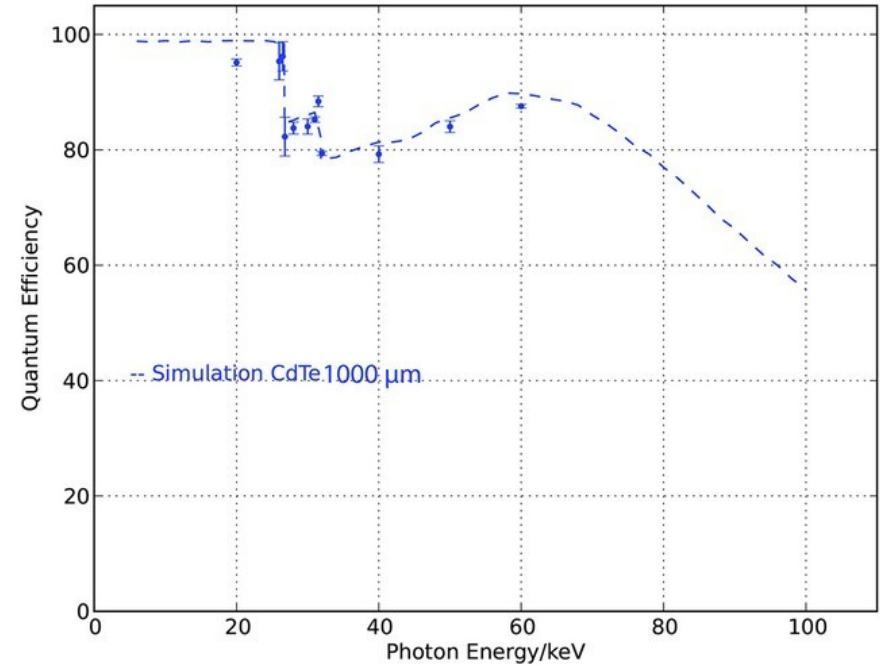
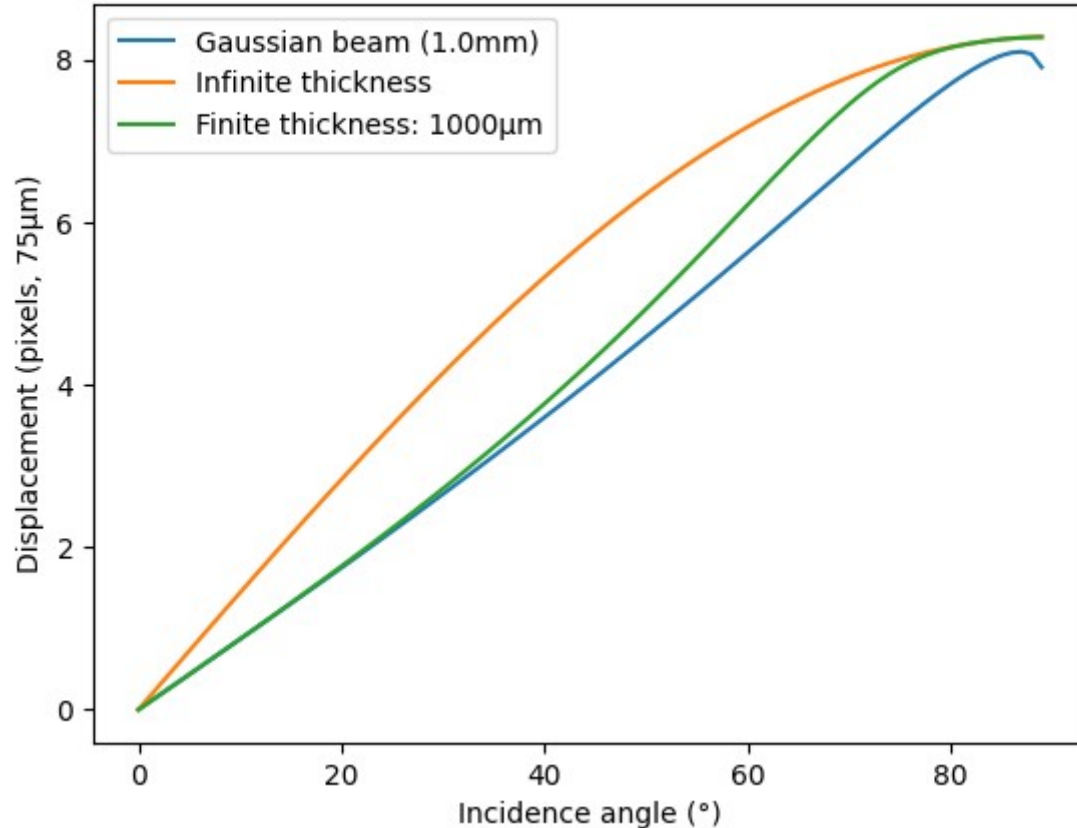
$$P(z) = \mu' \frac{e^{-\mu' z}}{1 - e^{-\mu' e}}$$

$$\mu' = \frac{\mu}{\cos(\alpha)}$$

$$\langle x \rangle = \frac{\sin(\alpha)}{\mu} \frac{1 - (1 + \mu' e) e^{-\mu' e}}{1 - e^{-\mu' e}}$$

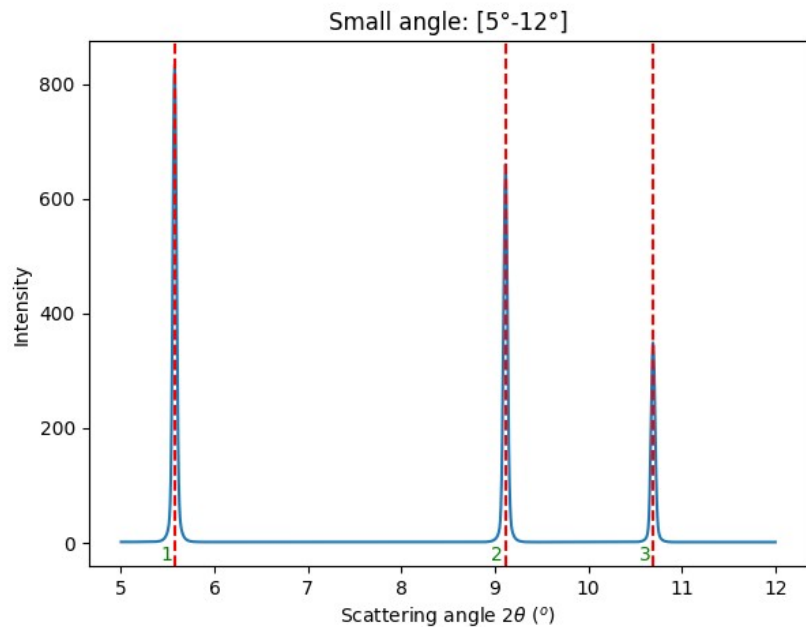
Comparison of models:

Displacement of beam due to parallax
CdTe @ 80keV, 80% efficiency

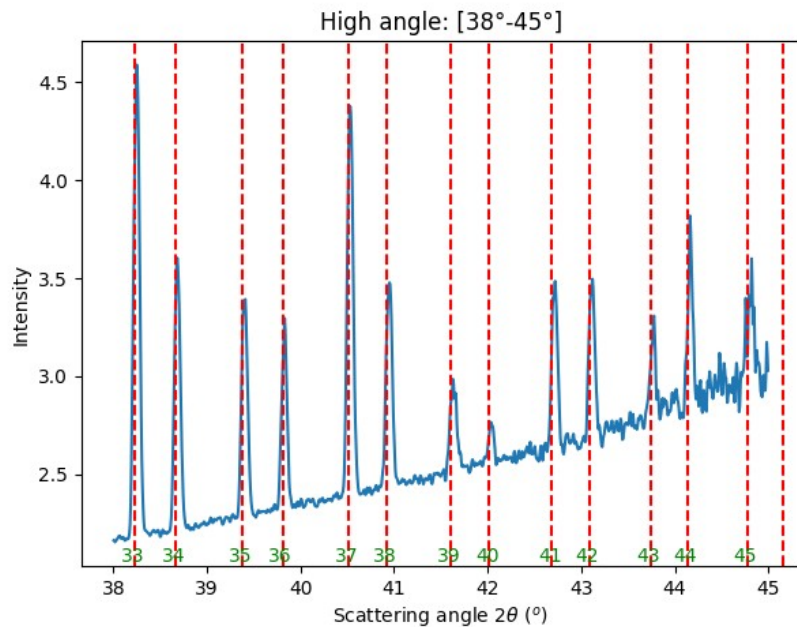


Dectris advertisement for CdTe Pilatus :
<http://dx.doi.org/10.1017/S0885715617001191>

Calibration with parallax correction



Data from ID11,
Eiger2 CdTe 4M, 750 μ m
Si sample @41keV
Calibration on ring 1 $\rightarrow$ 35



Possible from dev' version of pyFAI
(2025-09) and only in notebook version

Usage in pyFAI

```
In [1]: import pyFAI

In [2]: pyFAI.load({"detector": "eiger2_16M", "dist": 0.1, "wavelength": 8e-11})
Out[2]:
Detector Eiger2 16M      PixelSize= 75µm, 75µm      BottomRight(3)
Wavelength= 0.800000 Å
SampleDetDist= 1.000000e-01 m  PONI= 0.000000e+00, 0.000000e+00 m      rot1=0.000000 rot2=0.000000 rot3=0.000000 rad
DirectBeamDist= 100.000 mm      Center: x=0.000, y=0.000 pix      Tilt= 0.000° tiltPlanRotation= 0.000° λ= 0.800Å

In [3]: ai = pyFAI.load({"detector": "eiger2_16M", "dist": 0.1, "wavelength": 8e-11, "detector_config": {"sensor": {"material": "Si", "thickness": 1e-3}}})

In [4]: ai
Out[4]:
Detector Eiger2 16M      PixelSize= 75µm, 75µm      BottomRight(3) Si,1mm
Wavelength= 0.800000 Å Parallax: off
SampleDetDist= 1.000000e-01 m  PONI= 0.000000e+00, 0.000000e+00 m      rot1=0.000000 rot2=0.000000 rot3=0.000000 rad
DirectBeamDist= 100.000 mm      Center: x=0.000, y=0.000 pix      Tilt= 0.000° tiltPlanRotation= 0.000° λ= 0.800Å

In [5]: ref = ai.center_array(unit="r_m")

In [6]: ai.enable_parallax()

In [7]: ai
Out[7]:
Detector Eiger2 16M      PixelSize= 75µm, 75µm      BottomRight(3) Si,1mm
Wavelength= 0.800000 Å Parallax: µ= 22.719 cm-1
SampleDetDist= 1.000000e-01 m  PONI= 0.000000e+00, 0.000000e+00 m      rot1=0.000000 rot2=0.000000 rot3=0.000000 rad
DirectBeamDist= 100.000 mm      Center: x=0.000, y=0.000 pix      Tilt= 0.000° tiltPlanRotation= 0.000° λ= 0.800Å

In [8]: obt = ai.center_array(unit="r_m")

In [9]: (obt - ref) / ai.detector.pixel1
Out[9]:
array([[2.29946672e-03, 5.14176101e-03, 8.29083118e-03, ...,
        5.56211965e+00, 5.56228026e+00, 5.56244076e+00],
       [5.14176101e-03, 6.89839240e-03, 9.48092284e-03, ...,
        5.56211969e+00, 5.56228029e+00, 5.56244080e+00],
       [8.29083118e-03, 9.48092284e-03, 1.14972948e-02, ...,
        5.56211976e+00, 5.56228037e+00, 5.56244088e+00],
       ...,
       [5.59377994e+00, 5.59377997e+00, 5.59378003e+00, ...,
        5.72815575e+00, 5.72818822e+00, 5.72822068e+00],
       [5.59391643e+00, 5.59391646e+00, 5.59391652e+00, ...,
        5.72818990e+00, 5.72822235e+00, 5.72825480e+00],
       [5.59405283e+00, 5.59405286e+00, 5.59405292e+00, ...,
        5.72822403e+00, 5.72825647e+00, 5.72828890e+00]],
      shape=(4362, 4148))
```